

CHANDINI NEKKANTI

(508) 306-1856 | chandininekkanti23@gmail.com | [LinkedIn](#)

SUMMARY

Machine Learning Engineer focused on building agentic LLM systems, multi-agent pipelines, and RAG architectures for enterprise use cases. Experienced in orchestrating LLM workflows, fine-tuning open-source models, designing retrieval pipelines, and deploying scalable ML services. Strong background in model evaluation, observability, CI/CD, and scalable data pipelines across financial compliance and retail analytics environments.

TECHNICAL SKILLS

Agentic / Multi-Agent: LangChain, LlamaIndex, MCP, tool-use orchestration, ReAct patterns, multi-step reasoning, prompt engineering

ML / LLMs: PyTorch, TensorFlow, Scikit-learn, Llama 2/3, Mistral, LoRA/QLoRA, RAG, DPO/PPO, RLHF

Retrieval / GenAI: FAISS, BM25, cross-encoder reranking, RAGAS, vector search, MCP server integration

MLOps / Deployment: MLflow, Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Grafana, FastAPI

Inference Optimization: vLLM, NVIDIA Triton, batching, quantization, Flash Attention exposure

Data Engineering: Python, SQL, Spark, Kafka, Redis, gRPC, AWS SageMaker, AWS EKS, GCP Vertex AI, Azure ML

Evaluation: LLM evaluation, faithfulness/relevance scoring, drift monitoring, A/B testing, SHAP

EXPERIENCE

Morgan Stanley, USA

Machine Learning Engineer | Agentic AI & LLM Systems, July 2024 – Present

- Designed agentic LLM workflows using LangChain and LlamaIndex, enabling dynamic tool selection, multi-step reasoning, and chained document retrieval - reducing query resolution time for compliance analysts by 40%.
- Built agentic retrieval pipelines where LLMs dynamically route queries across FAISS vector search, BM25 keyword retrieval, and cross-encoder reranking stages - improving faithfulness/relevance scores by 23% and cutting manual document review time by 30%.
- Integrated MCP-compatible tool servers into LLM inference pipelines to enable structured tool-use and dynamic context retrieval, reducing prompt iteration cycles from 6 to 2 per workflow.
- Fine-tuned Mistral-7B on proprietary regulatory datasets using QLoRA (4-bit), reducing GPU memory usage by 75% versus full fine-tuning while maintaining performance on internal compliance QA benchmarks.
- Ran DPO fine-tuning experiments on curated financial QA preference pairs to improve agent response quality, reducing hallucinated regulatory citations by 35% as measured against internal compliance benchmarks.
- Helped optimize the inference stack using vLLM and NVIDIA Triton with INT8 quantization and dynamic batching, achieving 6.5x throughput improvement and 38% reduction in per-query GPU cost.
- Maintained and scaled Kubernetes-based inference services with autoscaling from 3 to 14 pods under peak load, sustaining p99 latency below 700ms at 180 RPS during traffic spikes.
- Orchestrated multi-model release workflows using MLflow registry and GitHub Actions CI/CD, enabling coordinated rollout and one-step rollback across 8 production models reducing release cycle from 2 days to 3 hours.
- Built observability dashboards using Prometheus and Grafana to monitor token latency, GPU utilization, drift signals, and production ML service health.

Virtual Infotech, India

Machine Learning (Data) Engineer | Mar 2020 – Dec 2022

- Built LSTM demand forecasting models on 4.2M+ retail transactions, reducing MAPE from 21% to 14% versus ARIMA baseline across 18 product categories providing inventory teams with a data-driven prediction engine.
- Fine-tuned BERT for customer feedback classification across 12 categories, achieving 0.89 F1 and automating tagging for 35K+ monthly feedback records, eliminating 18 hrs/week of manual effort.
- Applied GAN-based synthetic oversampling to improve model performance on imbalanced fraud detection datasets, increasing minority class recall for production risk models.
- Designed Spark and Kafka streaming pipelines to process daily retail event data in real time, replacing batch-based reporting workflows and reducing data freshness lag from 24 hours to near real-time.
- Improved inference API performance through request batching, Redis caching, and REST-to-gRPC migration for a customer-facing recommendation service, reducing average response latency significantly.

EDUCATION

Northeastern University

M.S. Data Analytics Engineering

Boston, MA

Jan 2023 – Dec 2024

V R Siddhartha Engineering College

B. Tech Electronics & Communication Engineering

India

Jun 2018 – May 2022